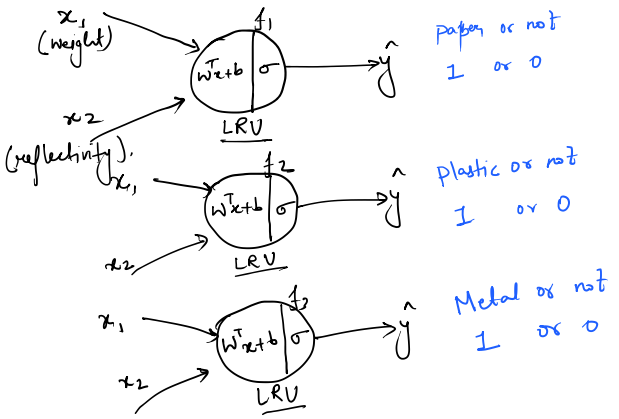
Forward & Backward propagation

Saturday, 19 September 2026 7:54 PM

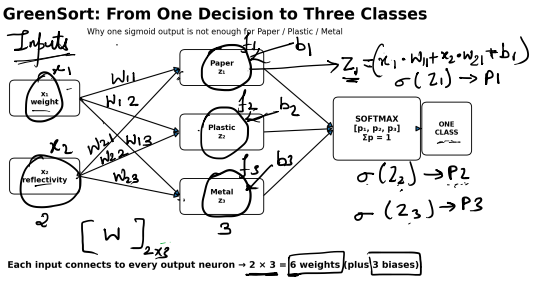
Green Sort

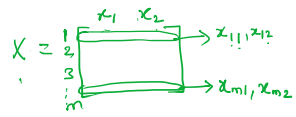
Design a solution to categories items into 3 classes: Paper, Plastic, Metal



$(W \cdot x + b)$ Z → Pre activated output.

$\sigma(z)$ P → activated output.



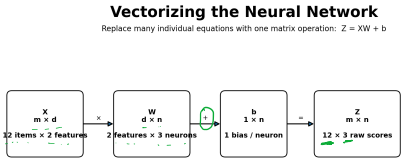


$W = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}$

$X \cdot W + b$

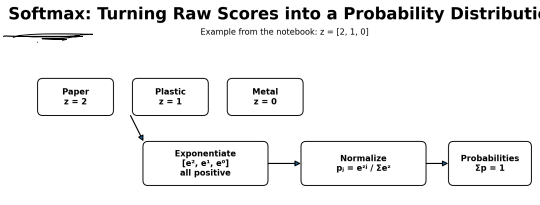
$(m \times 2) \cdot (2 \times 3) + [1 \times 3]$

$m \times 3$ (# of data points, no. of classes)



	Paper	Plastic	Metal
I_1	0.2	0.3	0.9
	0.6	0.3	0.9

> 0.5



Softmax(P_1, P_2, P_3)

Softmax $p_j = \dfrac{e^{z_j}}{\sum_{K=1}^{n} e^{z_K}}$ → Probabilities sum to 1.

I_1 $z(\text{Metal}) = 10$ ✓

I_1 $z(\text{Plastic}) = 5$

— $z(\text{Paper}) = 1$

$z_\text{Paper} = 2$ ✓

$z_\text{Plastic} = 1$

$z_\text{Metal} = 0$

y_i $\hat{y}_i$

I_i

class = 1 | class 0

$$\text{log-loss} = -\left[y_i \cdot \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i) \right]$$

$$1 \cdot \log(0.9) + (1-1) \cdot \log(\dots)$$

$I_i \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ 0.9

* Categorical Cross Entropy (CCE)

$I_j \rightarrow$ Metal

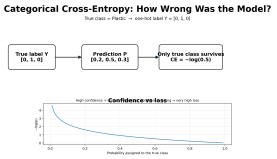
Item$_i$ → plastic

	Paper	Plastic	Metal
$y_{ij} \rightarrow$	0	1	0
	0	0	1
	↕	↕	↕
	0.3	0.1	0.6

Paper plastic metal.

I_i [0.1 0.7 0.2]

$-\log(0.7)$



$$CCE_i = -\sum_{j=1}^{K} y_{ij} \log(p_{ij})$$

$-\log(p_{i,\text{true class}})$. $-\log(0.6)$.

$y = [0 \;\; 1 \;\; 0]$

$p = [0.2 \;\; 0.5 \;\; 0.3]$

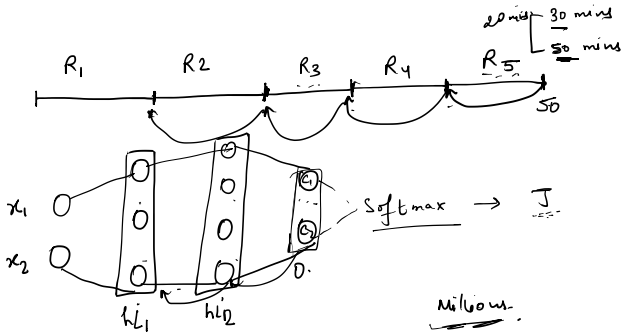
$-(0 \cdot \log(0.2) + 1 \cdot \log(0.5) + 0 \cdot \log(0.3))$

$-\log(0.5)$

* Forward Propogation.

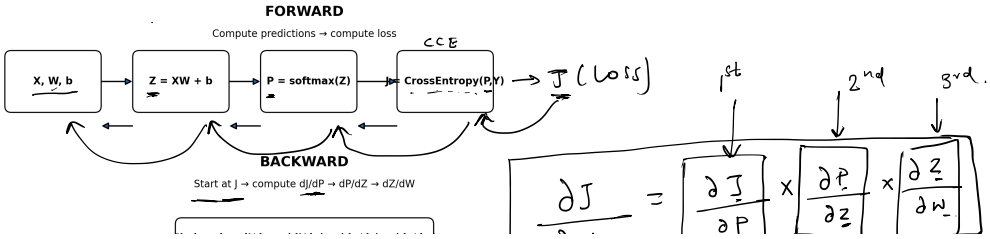
1) Compute Raw Scores = $Z \rightarrow XW + b$
2) Apply Softmax to Z to get the predicted probability P
3) Compare P against the true one hot labels Y to obtain the loss J.

W^*, b^* → Trainable parameters.



Forward Propagation vs Backward Propagation

Same computational graph — forward to compute loss, backward to assign responsibility



Chain rule: dJ/dw = (dJ/dp) × (dp/dz) × (dz/dw)

